

TIM 172B, Lecture #4 (1/15/26)

Agenda

1. Review SC strategy
2. Forecasting
3. HW #2 is posted on Canvas: please work on it.

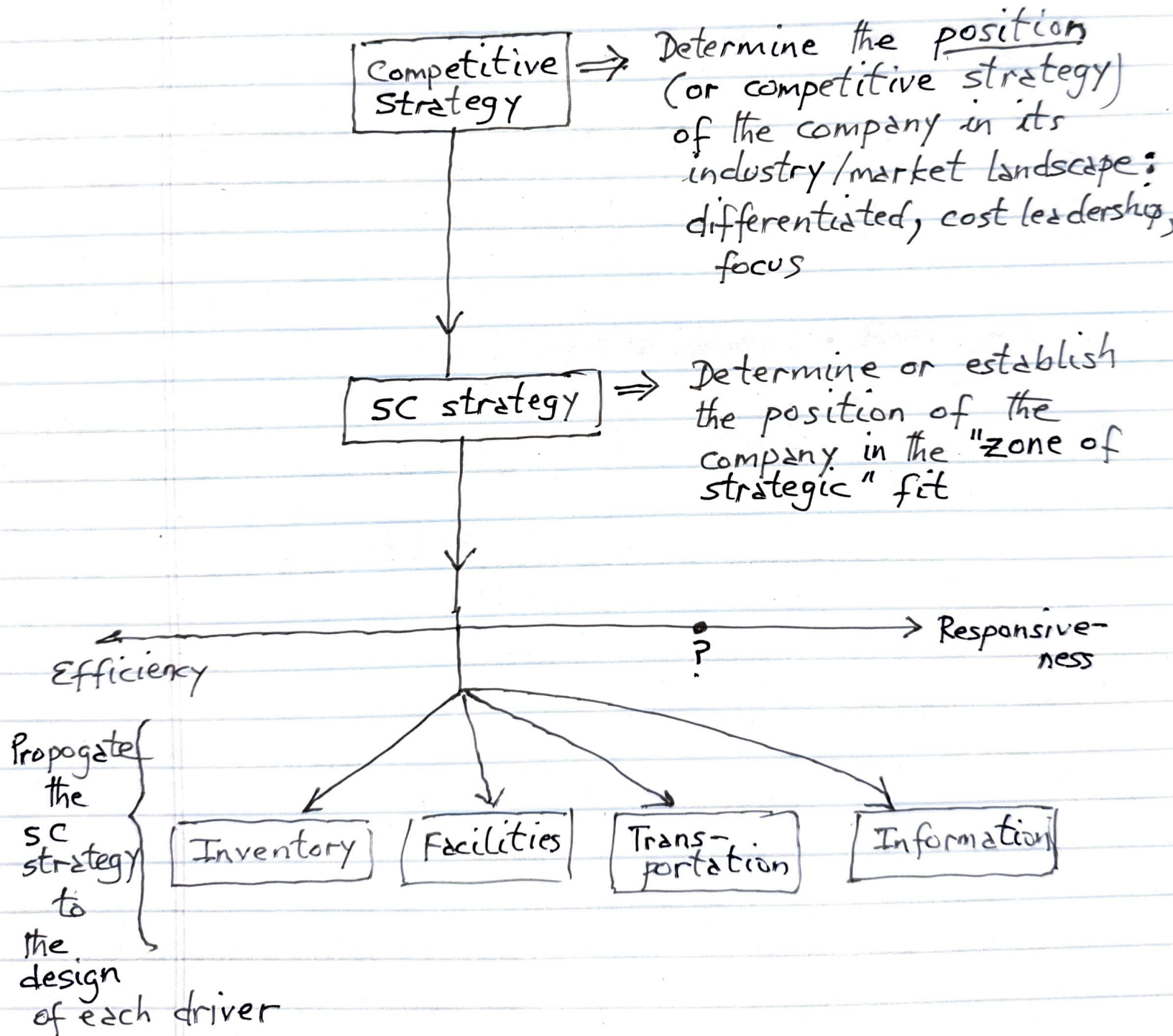
TIM 172Q Project Proposal: Read the "Apple SC" article (handout) on Canvas before/during working on your proposal.

4. Stages in Static Forecasting
(HW #2, Prob. 4: Static Forecasting ONLY)

1. Review SC strategy : Summary

Objective : Design the SC configuration or structure or network

Process :



See SCM textbook (Chapter on "SC Drivers")
for examples

Driver	Responsiveness	Efficiency
Inventory	• High inventory level • Wide range of products	• Low inventory levels • fewer
Facilities	Many locations close to the customer	• Few centralized locations to serve wide areas
Transportation	• Frequent shipments • Fast & flexible modes of transp.	• Few, larger shipments • slower, cheaper modes of transp.
Information	• collect & share accurate data in a timely fashion	• reduce cost of information

2. (Demand) Forecasting Fundamentals

1. Read & use the chapter on "Demand Forecasting in a SC" in the text.
2. HW # 2, Prob. 4
⇒ do the worked example in the Demand Forecasting chapter in the text
- Static Forecasting ONLY.

Fundamentals:

1. In order to plan every aspect of the SC, the company needs to forecast demand accurately.

Example : DELL : Why does Dell need to forecast demand

- Dell needs to anticipate customer demand in order to order sub-systems (e.g., microprocessor,) from its suppliers.
- Production managers must plan for the required capacity in the manf. plant
- ...

Dell forecasts demand using "Time Series Analysis", which uses historical demand data to make forecasts (or predictions) about future demand.

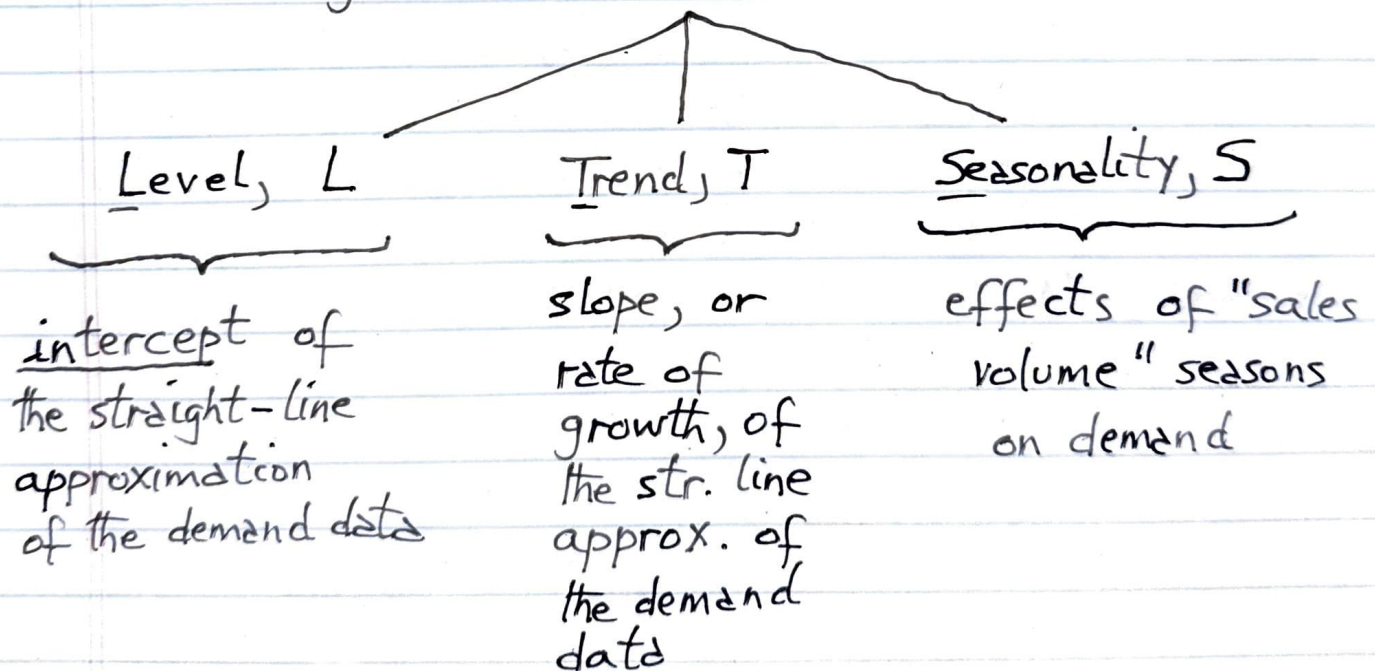
2. Observed (or actual) demand, O

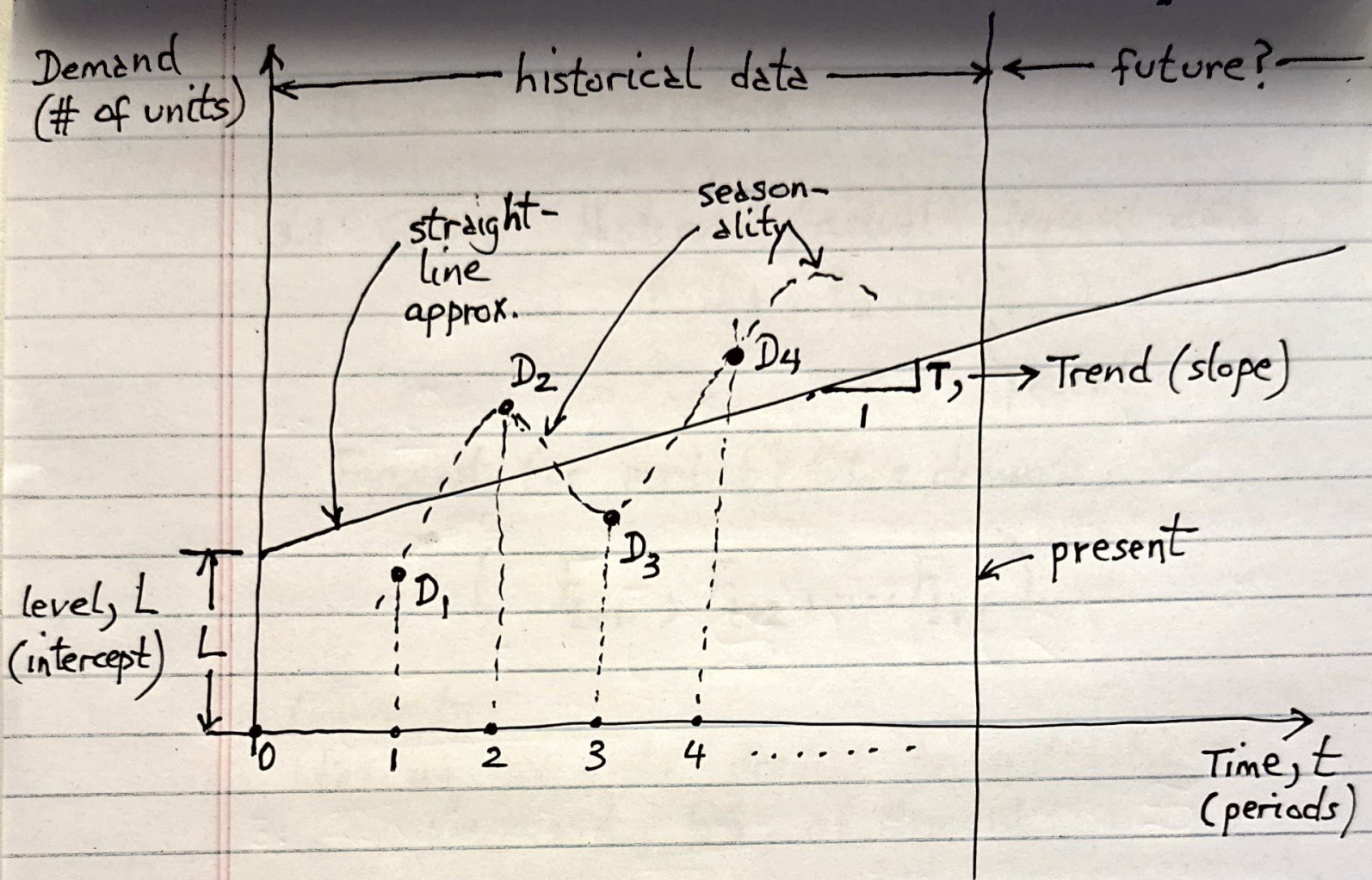
$$O = \text{systematic component} + \text{Random comp.}$$

$$O = (S) + (R)$$

3. Systematic component of demand, S
 $\equiv$ expected value of demand,

in general has 3 components





(a) $D_1, D_2, \dots$ actual (historical) data pts.

(b) When we fit a straight-line to these data pts., the str. line has a Y-axis intercept, called the level, L ; and a slope, or rate of growth, called the trend, T

"High-school" formula for a str. line, $y = mx + b$
↑ ↑
slope intercept
 $= b + xm$

For a discrete str. line shown above,

$$\bar{D}_t = L + tT; \quad t = 0, 1, 2, \dots$$

↑ ↑
demand at Trend
time t (slope)
 Level
 (intercept)

3. Demand Forecasting

3.1 Given : Historical (actual) demand data,
 $(D_1, D_2, \dots, D_t)$

 present

Forecast (or predict) future demands,

$$(F_{t+1}, F_{t+2}, \dots, F_{t+l})$$

Comments:

(i) we can only predict the systematic component, S , of demand.

(ii) the random component of demand, R , cannot be predicted, but can be bounded or estimated using the forecasting error.

3.2. Forecasting methods

Static Forecasting

Assumption : L , T , and seasonality, S , do not vary with time.

Adaptive Forecasting:
update the estimates
of L , T , and S , as
new demand data
becomes available.

3.3 Preliminary ideas for forecasting

(i) Plotting : always plot the data, before, during, and after performing demand forecasting
 $\Rightarrow$ (1) to understand the data, and (2) later, to see how well your forecast predicts the data

(ii) Regression : determine the straight line that "best" fits the data
("best" in a "least-squares" sense)

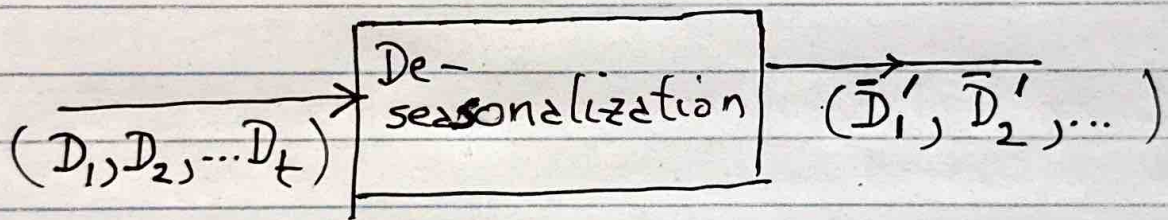
(iii) Static forecasting is important for 2 reasons:
(i) understand/learn how forecasting works
(ii) as a "seed" for initiating adaptive forecasting.

4. Stages in Static Forecasting

There are 4 important stages in static forecasting

(i) De-seasonalization: The actual demand data pts. D_i , ($i=1,2,\dots,t$) are first de-seasonalized to obtain the de-seasonalized data pts, $\bar{D}'_i$, ($i=1,2,\dots,t$)

[De-seasonalization "flattens" or compresses the data



(ii) Regression.